# Concentric Tube Robot — Mathematical Description

This document describes the geometry and kinematics behind the concentric tube robot (CTR) generated by `ctr_generator.py`.

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## 1. Tube Parameters

Two coaxial tubes, all dimensions in metres after the conversion factor $\kappa = 10^{-3}$:

| Tube | Length $L$ | Inner radius $r_i$ | Outer radius $r_o$ | Curvature radius $R$ |
|---|---|---|---|---|
| Outer | 30 mm | 2.0 mm | 2.5 mm | $\infty$ (straight) |
| Inner | 30 mm | 1.55 mm | 1.80 mm | 50 mm |

For each tube the radii are derived from the diameters as $r = d/2$.

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## 2. Centerline Parameterization

A centerline is a discretised curve $\mathbf{c} : [0, L] \to \mathbb{R}^3$ sampled at $N$ points $\{\mathbf{p}_k\}_{k=0}^{N-1}$.

### 2.1 Straight tube (outer)

$$\mathbf{c}(s) = \begin{pmatrix} 0 \\ 0 \\ s \end{pmatrix}, \qquad s \in [0, L]$$

with $N = 40$ samples uniformly in $s$.

### 2.2 Curved tube (inner)

The pre-curved inner tube follows a planar circular arc of radius $R$ in the $xz$-plane. Parameterising by angle $\varphi$:

$$\mathbf{c}(\varphi) = \begin{pmatrix} R\,(1 - \cos\varphi) \\ 0 \\ R \sin\varphi \end{pmatrix}, \qquad \varphi \in [0, \theta]$$

where the **total subtended angle** is

$$\theta = \frac{L}{R}.$$

For $L = 0.030\,\mathrm{m}$ and $R = 0.050\,\mathrm{m}$, $\theta = 0.6\,\mathrm{rad} \approx 34.4^\circ$. Sampling uses $N = 120$ angles uniformly in $[0, \theta]$.

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## 3. Local Frame (Frenet-Like)

To sweep a circular cross-section along the centerline, an orthonormal frame $(\mathbf{t}_k, \mathbf{n}_{1,k}, \mathbf{n}_{2,k})$ is computed at every sample point.

### 3.1 Tangent

Using central differences for interior points and one-sided differences at the endpoints:

$$\tilde{\mathbf{t}}_k = \begin{cases} \mathbf{p}_1 - \mathbf{p}_0 & k = 0 \\ \mathbf{p}_{k+1} - \mathbf{p}_{k-1} & 0 < k < N-1 \\ \mathbf{p}_{N-1} - \mathbf{p}_{N-2} & k = N-1 \end{cases}, \qquad \mathbf{t}_k = \frac{\tilde{\mathbf{t}}_k}{\|\tilde{\mathbf{t}}_k\|}$$

### 3.2 Normals

A reference vector $\hat{\mathbf{e}}_{\text{ref}}$ is chosen to seed the cross product. To avoid degeneracy when $\mathbf{t}_k$ is nearly parallel to the reference,

$$\hat{\mathbf{e}}_{\text{ref}}^{(k)} = \begin{cases} \hat{\mathbf{y}} & |\mathbf{t}_k \cdot \hat{\mathbf{y}}| \le 0.95 \\ \hat{\mathbf{x}} & \text{otherwise} \end{cases}$$

The two surface normals are then

$$\mathbf{n}_{1,k} = \frac{\hat{\mathbf{e}}_{\text{ref}}^{(k)} \times \mathbf{t}_k}{\|\hat{\mathbf{e}}_{\text{ref}}^{(k)} \times \mathbf{t}_k\|}, \qquad \mathbf{n}_{2,k} = \mathbf{t}_k \times \mathbf{n}_{1,k}$$

By construction $(\mathbf{t}_k, \mathbf{n}_{1,k}, \mathbf{n}_{2,k})$ is right-handed and orthonormal.

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## 4. Hollow Tube Mesh

A watertight hollow cylindrical shell is generated by sweeping two concentric rings of $S = 72$ vertices along the centerline.

### 4.1 Ring vertex placement

At sample $k$, the angular subdivisions of the cross-section are

$$\alpha_j = \frac{2\pi j}{S}, \qquad j = 0, 1, \ldots, S-1.$$

Outer and inner ring vertices are

$$\mathbf{v}_{k,j}^{\text{out}} = \mathbf{p}_k + r_o\big(\cos\alpha_j\,\mathbf{n}_{1,k} + \sin\alpha_j\,\mathbf{n}_{2,k}\big)$$

$$\mathbf{v}_{k,j}^{\text{in}} = \mathbf{p}_k + r_i\big(\cos\alpha_j\,\mathbf{n}_{1,k} + \sin\alpha_j\,\mathbf{n}_{2,k}\big)$$

The total vertex count is $2NS$.

### 4.2 Triangulation

Adjacent rings are connected by quadrilaterals split into two triangles. For the outer wall, with $j' = (j+1) \bmod S$:

$$T_1 = (\mathbf{v}_{k,j}^{\text{out}},\ \mathbf{v}_{k+1,j}^{\text{out}},\ \mathbf{v}_{k,j'}^{\text{out}}), \quad T_2 = (\mathbf{v}_{k,j'}^{\text{out}},\ \mathbf{v}_{k+1,j}^{\text{out}},\ \mathbf{v}_{k+1,j'}^{\text{out}})$$

The inner wall uses the **reversed winding order** so its normals point inward (i.e. into the tube bore, opposite the outer wall).

### 4.3 End caps

To make the shell watertight, annular caps close the ends by bridging the outer and inner rings at $k = 0$ and $k = N-1$ with the same two-triangle pattern.

### 4.4 Triangle normals

For each triangle $(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2)$ written to the STL,

$$\mathbf{n} = \frac{(\mathbf{v}_1 - \mathbf{v}_0) \times (\mathbf{v}_2 - \mathbf{v}_0)}{\|(\mathbf{v}_1 - \mathbf{v}_0) \times (\mathbf{v}_2 - \mathbf{v}_0)\|}$$

Degenerate triangles ($\|\cdot\| < 10^{-12}$) fall back to $\mathbf{n} = \hat{\mathbf{z}}$.

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## 5. Kinematic Model

### 5.1 Configuration vector

$$\mathbf{q} = \left(x_b,\ y_b,\ z_b,\ d,\ \psi,\ \beta_1,\ \beta_2,\ \ldots,\ \beta_M\right)^\top \in \mathbb{R}^{5+M}$$

where

- $(x_b, y_b, z_b)$ — base translation in the world frame,
- $d \in [0, L]$ — insertion depth of the inner tube,
- $\psi \in [-\pi, \pi]$ — axial rotation of the inner tube,
- $\beta_i$ — bending angle of segment $i$ for $i = 1, \ldots, M$.

The number of inner-tube segments is $M = 15$.

### 5.2 Joint limits

| Joint | Lower | Upper |
|---|---|---|
| $x_b, y_b, z_b$ | $-0.05\,\mathrm{m}$ | $+0.05\,\mathrm{m}$ |
| $d$ | $0$ | $0.03\,\mathrm{m}$ |
| $\psi$ | $-\pi$ | $+\pi$ |
| $\beta_i$ | $-\Delta\beta$ | $+\Delta\beta$ |

### 5.3 Per-segment bend angle

The inner tube's total intrinsic curvature angle $\theta = L/R$ is distributed uniformly across $M$ revolute joints (each rotating about the local $\hat{\mathbf{y}}$-axis):

$$\Delta\beta = \frac{\theta}{M} = \frac{L}{M\,R}$$

For the default parameters,

$$\Delta\beta = \frac{0.030}{15 \cdot 0.050} = 0.04\,\mathrm{rad} \approx 2.29^\circ \quad \text{per joint.}$$

Each segment has length

$$\ell_s = \frac{L}{M} = 0.002\,\mathrm{m} = 2\,\mathrm{mm}.$$

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## 6. Insertion-Driven Bending

The inner tube is modelled as initially straight; as it is inserted by a depth $d$, segments **emerge** from the outer tube tip and progressively rotate to recover the natural curve. This is captured by an activation function per joint.

### 6.1 Activation per joint

For joint $i \in \{1, \ldots, M\}$, the distance from the inner tube tip is

$$\delta_i = (M - i)\,\ell_s.$$

The activation $a_i \in [0, 1]$ is

$$a_i(d) = \mathrm{clip}\left(\frac{d - \delta_i}{\ell_s},\ 0,\ 1\right)$$

with $\mathrm{clip}(x, a, b) = \min(\max(x, a), b)$.

### 6.2 Bend angle as a function of insertion

$$\beta_i(d) = a_i(d) \cdot \Delta\beta$$

This produces a piecewise-linear ramp: each joint stays at 0 until its segment begins to emerge, ramps linearly to $\Delta\beta$ over a stroke of $\ell_s$, and saturates thereafter. The tip joint ($i = M$) activates first; the basal joint ($i = 1$) activates last.

### 6.3 Total deflection

The total curve angle accumulated along the emerged portion is

$$\Theta(d) = \sum_{i=1}^{M} \beta_i(d) = \Delta\beta \sum_{i=1}^{M} a_i(d).$$

When fully inserted ($d = L$), all $a_i = 1$ and $\Theta(L) = M\,\Delta\beta = \theta = L/R$, recovering the natural curvature angle.

## 7. Forward Kinematics Sketch

Each joint contributes a homogeneous transform $T_i \in \mathrm{SE}(3)$. The world-to-tip transform is

$$T_{\text{tip}}(\mathbf{q}) = T_{x_b}\, T_{y_b}\, T_{z_b}\, T_{\text{outer}}\, T_d\, T_\psi \prod_{i=1}^{M} T_{\beta_i}\, T_{\ell_s}$$

with the per-segment factor

$$T_{\beta_i}\, T_{\ell_s} = \underbrace{\begin{pmatrix} R_y(\beta_i) & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{pmatrix}}_{\text{rotate about } \hat{\mathbf{y}}} \underbrace{\begin{pmatrix} I_3 & \ell_s \hat{\mathbf{z}} \\ \mathbf{0}^\top & 1 \end{pmatrix}}_{\text{translate along local } \hat{\mathbf{z}}}$$

In the limit $M \to \infty$ with $\Delta\beta = \theta/M$, the chain converges to the continuous arc of radius $R$ subtending angle $\theta$, exactly the centerline of §2.2.

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## 8. Summary of Numerical Values

| Symbol | Value | Description |
|---|---|---|
| $L$ | $0.030\,\mathrm{m}$ | Tube length |
| $R$ | $0.050\,\mathrm{m}$ | Inner tube curvature radius |
| $\theta = L/R$ | $0.600\,\mathrm{rad}$ | Total curve angle |
| $M$ | $15$ | Discrete bend joints |
| $\ell_s = L/M$ | $0.002\,\mathrm{m}$ | Segment length |
| $\Delta\beta = \theta/M$ | $0.040\,\mathrm{rad}$ | Bend per joint |
| $S$ | $72$ | Cross-section subdivisions |
| $N_{\text{outer}}$ | $40$ | Centerline samples (outer) |
| $N_{\text{inner}}$ | $120$ | Centerline samples (inner segment, before discretisation) |